

INDUSTRIAL WELDING ROBOT

Complete Technical Reference Manual

Covers Mechanical, Electrical, Control, Kinematics,
Sensors, Power, Communication & Manufacturing

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CHAPTER 00

Overview

Introduction to Industrial Welding Robots

Industrial welding robots are programmable, multi-axis manipulators designed to perform arc welding, spot welding, laser welding, and allied joining processes with high speed, repeatability, and precision. Modern systems integrate advanced sensors, real-time controllers, and AI-driven seam-tracking to deliver weld quality that surpasses manual operations while dramatically reducing cycle time and operator fatigue.

The welding robot described in this manual is a 6-DOF articulated arm with a dedicated welding power source, torch cooling system, wire-feed unit, and a safety-rated collaborative workspace envelope. It is designed for integration into automated production lines in automotive, heavy equipment, shipbuilding, and structural-steel fabrication sectors.

Key Features

- Six-axis articulated arm — reach 1 850 mm, payload 20 kg
- Integrated arc-length control via arc-voltage feedback
- Seam-tracking with laser vision sensor (± 0.1 mm accuracy)
- Touch-sensing for weld-start location detection
- EtherCAT real-time bus — 1 ms cycle time
- Compatible with MIG/MAG, TIG, Plasma, and Laser processes
- Rated IP54; optional IP67 for harsh environments

Working Principle

The robot receives a welding program (torch path + process parameters) from the teach pendant or offline programming software. The motion controller decomposes the Cartesian path into joint-space trajectories using inverse kinematics. Each joint is driven by a servo motor through a harmonic-drive gearbox, providing zero-backlash motion essential for weld bead consistency.

During welding, the arc sensor continuously monitors arc voltage and current. Deviations from set-point trigger real-time corrections to torch height and lateral position. The laser vision unit (mounted ahead of the torch) detects joint geometry changes and updates the path adaptively, compensating for part distortion and fixture tolerances.

Parameter	Value	Unit
Degrees of Freedom	6	axes
Maximum Reach	1 850	mm
Wrist Payload	20	kg
Repeatability	± 0.04	mm
Maximum TCP Speed	2 000	mm/s
Welding Current Range	30 – 500	A
Duty Cycle	100 % at 350 A	—
Wire Feed Speed	1 – 25	m/min
Power Consumption (peak)	15	kVA

Applications

- Automotive body-in-white — door panels, floor pan, chassis rails
- Heavy equipment — excavator booms, loader arms, frame assemblies
- Shipbuilding — longitudinal stiffeners, bracket welding
- Pressure vessels & piping — circumferential and longitudinal seams
- Structural steel — column-to-beam connections, plate girders
- Aerospace sub-structures — aluminium skin stringer welding
- Agricultural machinery — implement frames, lifting attachments

CHAPTER 01

Mechanical System

Frame & Structural Design

The robot base is a rigid cast-iron pedestal bolted to the shop floor via anchor bolts torqued to 120 N·m. The arm links are fabricated from high-strength aluminium alloy (AA 7075-T6) to reduce moving inertia while maintaining stiffness. Finite-element analysis confirms a natural frequency above 30 Hz for all structural modes, preventing resonance at normal operating speeds.

Component	Material	Mass (kg)	Remarks
Base Pedestal	Cast Iron GG25	85	Floor-mounted
Link 1 (Shoulder)	AA 7075-T6	12.4	Rotary, J1 axis
Link 2 (Upper arm)	AA 7075-T6	9.8	J2 axis
Link 3 (Forearm)	AA 7075-T6	6.2	J3 axis
Wrist Assembly	Steel + AA	3.1	J4–J6 axes
Torch Bracket	SS 316L	0.8	Water-cooled

Harmonic Drive Gearboxes

Each axis uses a zero-backlash harmonic-drive (strain-wave) gearbox. The flex-spline, wave generator, and circular spline arrangement delivers gear ratios from 1:80 (shoulder) to 1:160 (wrist), providing high torque density and compact form factor. Lubrication is lifetime grease fill; no field re-lubrication required for 20 000 hours.

Welding Torch Assembly

The torch mount accepts standard 22 mm euro-fitting torches. A pneumatic quick-change coupling (crash protection) releases at 80 N lateral force, preventing torch damage on collision. A water-cooled torch body with 4 L/min flow is standard; air-cooled torches may be fitted for currents below 200 A.

- Torch tilt: ±180° continuous rotation on J6
- Wire conduit: 3.0 m spiral-steel liner, 1.0 / 1.2 mm diameter
- Nozzle-cleaning station: integrated reamer + anti-spatter spray
- Wire cutter: pneumatic, ±0.5 mm wire stick-out control

End-Effector & Tool Changer

An optional automatic tool changer (ATC) at the wrist flange allows the robot to swap between welding torch, grinding wheel, and inspection probe within a single program. The ATC uses a pneumatic locking ring and spring-loaded electrical/fluid pass-through. Tool weight limit: 12 kg; repeatability: ±0.01 mm.

CHAPTER 02

Motors

Servo Motor Overview

All six robot axes use brushless AC servo motors (PMSM — Permanent Magnet Synchronous Motors) with integrated 23-bit absolute encoders. The absolute encoder eliminates homing cycles after power-up, a critical safety and productivity feature. Stator windings are rated Class H (180 °C), suitable for continuous high-duty operation.

Axis	Motor Model	Rated Torque (N·m)	Rated Speed (rpm)	Encoder Resolution
J1 Waist	SGM7J-15A	14.3	3 000	23-bit abs
J2 Shoulder	SGM7J-15A	14.3	3 000	23-bit abs
J3 Elbow	SGM7J-08A	7.6	3 000	23-bit abs
J4 Wrist Roll	SGM7J-04A	3.8	3 000	23-bit abs
J5 Wrist Pitch	SGM7J-04A	3.8	3 000	23-bit abs
J6 Wrist Yaw	SGM7J-02A	1.9	5 000	23-bit abs

Wire Feed Motor

A dedicated 24 V DC brushless motor drives the 4-roll wire feeder. Encoder feedback maintains wire feed speed within ±0.5 % of set-point across 1 – 25 m/min. The drive electronics are integrated into the welding power source cabinet.

Motor Thermal Management

Each servo motor includes a KTY84 thermistor embedded in the stator. The drive firmware monitors temperature and reduces current limit (I_t derating) above 120 °C, preventing insulation damage. Forced-air cooling fins increase continuous rated torque by 15 % compared to natural-convection cooling.

CHAPTER 03

Motor Drivers

Drive Architecture

Six servo drives (one per axis) are mounted in the robot controller cabinet. Each drive is a three-phase PWM inverter based on IGBT bridge (600 V, 50 A). The DC bus is shared across all drives, allowing regenerative braking energy from decelerating axes to power accelerating axes, reducing peak grid demand by up to 30 %.

Specification	Value
Drive supply voltage	400 V AC ±10 %, 3-phase 50/60 Hz
DC bus voltage	560 V DC nominal
PWM switching frequency	8 kHz (configurable 4–16 kHz)
Current loop bandwidth	2 kHz
Velocity loop bandwidth	300 Hz
Position loop bandwidth	100 Hz
EtherCAT cycle time	1 ms (125 µs optional)
Overload capacity	300 % for 3 s
Protection class	IP20 (cabinet-mounted)

Protection Circuits

- Over-current protection — hardware trip < 1 µs via gate driver
- Over-voltage crowbar — activates at 700 V DC bus
- Under-voltage lockout — disables PWM below 450 V DC
- Short-circuit protection — current sensing on each phase
- Ground fault detection — isolated monitoring GFCI module
- Safe Torque Off (STO) — PLe/Cat 4 per IEC 62061

PWM Control Strategy

Space-Vector PWM (SVPWM) minimises harmonic content and reduces switching losses compared to sinusoidal PWM. Field-Oriented Control (FOC) decouples d-axis (flux) and q-axis (torque) currents, enabling linear torque response independent of rotor position. Auto-tuning routines identify motor parameters (resistance, inductance, back-EMF) during commissioning.

CHAPTER 04

Sensors

Arc Sensing

Arc-voltage and welding-current signals are fed back to the welding controller at 10 kHz sample rate. Voltage is measured at the torch (not at the power source) to eliminate cable resistance error. The controller uses an adaptive PI algorithm to maintain the set arc length by adjusting wire feed speed.

Laser Vision Seam Tracking

A structured-light laser triangulation sensor (650 nm, Class 2) is mounted 80 mm ahead of the torch. It projects a line pattern across the joint and a CMOS camera images the reflected stripe. A DSP processes the image at 500 Hz to extract joint position and geometry.

Sensor Parameter	Specification
Measurement range (Z)	60 – 120 mm from sensor face
Lateral field of view	±25 mm
Z accuracy	±0.05 mm (1σ)
Lateral accuracy	±0.1 mm (1σ)
Output rate	500 Hz
Seam types detected	Butt, lap, fillet, V-groove, pipe saddle
Ingress protection	IP67 with protective glass

Touch Sensing

Before each weld start, the robot probes the joint using touch sensing (wire tip as electrode, workpiece as return). Contact detection at < 1 mA trigger current identifies exact part position, compensating for fixture tolerances up to ±5 mm. Up to three touch-sense points can be programmed per weld seam.

Force / Torque Sensor

A 6-axis F/T sensor (Fx, Fy, Fz, Tx, Ty, Tz) at the wrist enables force-controlled grinding, polishing, and assembly tasks. Range: ±500 N / ±20 N·m; resolution 0.1 N / 0.005 N·m. The sensor is fully shielded against welding EMI.

Safety Sensors

- Axis-mounted dual-channel encoders for safe speed/position monitoring
- Collision detection via motor current monitoring (< 5 ms response)
- Area scanner (SICK S300) — safety-rated light curtain around cell
- Door interlock switches — category 4, PLe
- Emergency stop buttons on teach pendant, control panel, and cell fence

CHAPTER 05

Control System

Controller Architecture

The robot controller is a real-time multi-core PC (Intel Xeon, 8 cores) running a RTOS (EtherLab / Xenomai). The motion kernel executes at 1 ms cycle time, handling trajectory interpolation, inverse kinematics, and servo command generation. A non-real-time process handles HMI, program management, and connectivity.

Module	Function	Bus Interface
Motion Controller	Trajectory + IK + servo commands	EtherCAT master
Welding Controller	Arc parameters + seam track	EtherCAT slave / CAN
Safety Controller	STO, SLS, SLP monitoring	FSoE over EtherCAT
HMI PC	Program editor, monitoring, logs	Gigabit Ethernet
I/O Module	Digital/Analog I/O for cell	EtherCAT I/O
Vision PC	Seam tracking image processing	GigE Vision

Programming Environment

- Teach pendant: 10.1" colour touchscreen, IP65, wireless option
- Offline programming: RobotStudio / WorkVisual compatible
- Language: proprietary Rapid-like script + G-code import
- Simulation: 3D collision checking, cycle-time estimation
- OPC-UA server for MES/SCADA integration

HMI & Diagnostics

The web-based HMI (accessible from any browser on the plant network) displays real-time weld parameters, robot joint states, alarm history, and maintenance counters. Trending graphs allow quality engineers to correlate arc data with weld bead geometry. Automatic alarm suppression during programmed process interruptions reduces nuisance alerts.

CHAPTER 06

Power System

Welding Power Source

The welding power source is an inverter-based constant-voltage / constant-current unit with 400 V 3-phase input. IGBT full-bridge topology achieves > 85 % efficiency. Weld programs can switch between CV (MIG/MAG), CC (TIG), and pulsed modes under software control within 10 ms.

Parameter	Specification
Input supply	400 V AC ±15 %, 3-phase, 50/60 Hz
Output current range	30 – 500 A DC
Output voltage range	12 – 44 V DC
Duty cycle	100 % @ 350 A, 60 % @ 500 A
Open-circuit voltage	68 V max
Wire feed motor supply	24 V DC, 5 A
Cooling	Forced air; thermal protection 85 °C
Standards	IEC 60974-1, -10 (EMC Class A)

Robot Controller Power Supply

A 24 V DC SMPS (redundant, hot-swap) powers all control electronics. A 48 V DC bus supplies regenerative servo drives. An integrated UPS (Li-ion, 5 min hold-up) enables a controlled retract move on mains failure, preventing torch crash into the workpiece.

Grounding & Shielding

Welding return current must flow through the dedicated welding cable, NOT through robot structure or signal cables. A welding return connection point is provided on the robot base. All signal cables are individually shielded; shields are single-point grounded at the controller cabinet to prevent ground-loop interference from the welding arc.

Note: Failure to follow the grounding guidelines may cause encoder communication errors and premature bearing failure due to shaft currents.

CHAPTER 07

Communication

Real-Time Fieldbus: EtherCAT

EtherCAT (IEC 61158) is used for all real-time axis and I/O communication. The master runs on the motion controller; servo drives, I/O modules, and the welding controller are slaves. Distributed Clock (DC) synchronisation aligns all devices to $\pm 1 \mu\text{s}$, ensuring tight multi-axis coordination.

Industrial Ethernet: OPC-UA

OPC-UA (IEC 62541) provides platform-independent connectivity to MES, SCADA, and cloud systems. The robot exposes a standardised information model including weld job status, parameter setpoints, actual values, alarm states, and maintenance counters. TLS 1.3 encryption and certificate-based authentication are mandatory for all OPC-UA connections.

Safety Bus: FSoE

Functional Safety over EtherCAT (FSoE) carries safety-rated signals (E-stop, STO, door interlock, area scanner) over the same physical EtherCAT cable. FSoE achieves SIL 3 / PLe with a residual error probability < 10^{-9} /h.

Protocol	Layer	Use Case	Cycle Time
EtherCAT	L2 Ethernet	Servo + I/O control	1 ms
FSoE	L2 / EtherCAT	Safety signals	8 ms (SIL3)
OPC-UA	TCP/IP	MES / SCADA / cloud	100 ms – 1 s
GigE Vision	UDP/IP	Vision camera data	2 ms per frame
PROFINET IRT	L2 Ethernet	Optional PLC link	1 ms
Modbus TCP	TCP/IP	Legacy PLC interface	100 ms

CHAPTER 08

Kinematics

Denavit-Hartenberg Parameters

The robot kinematic model uses the modified DH convention. Six homogeneous transformation matrices (T1...T6) are composed to yield the end-effector pose in the base frame. Table 08-1 lists the DH parameters for the nominal geometry.

Joint i	a (mm)	alpha (°)	d (mm)	theta offset (°)	Range (°)
1	0	0	400	0	±170
2	650	-90	0	-90	-65 / +145
3	150	0	0	0	-70 / +190
4	0	90	700	0	±190
5	0	-90	0	0	±120
6	0	90	100	0	±360

Forward Kinematics

The forward kinematic function maps joint angles (q1...q6) to TCP pose (x, y, z, Rx, Ry, Rz). It is computed in closed form by chaining the six DH transformation matrices. Execution time on the motion controller is < 10 µs, allowing real-time evaluation at every 1 ms servo cycle.

Inverse Kinematics

A closed-form IK solver uses the decoupled wrist approach: the wrist-centre position determines J1–J3 (position sub-problem), then the wrist Euler angles determine J4–J6 (orientation sub-problem). Up to eight solution configurations exist; the solver selects the nearest configuration that avoids joint limits and singularities.

Workspace Analysis

The reachable workspace is a torus-shaped volume centred on the robot base. The dexterous workspace (all TCP orientations reachable) occupies approximately 70 % of the reachable volume. Singularity zones (shoulder, elbow, and wrist singularities) are software-restricted; the controller automatically detects proximity and limits velocity to prevent joint-speed divergence.

- Maximum horizontal reach from base: 1 850 mm
- Vertical reach (above base): 2 320 mm
- Vertical reach (below base): 480 mm
- Wrist singularity avoidance threshold: $J5 \leq \pm 1^\circ$

CHAPTER 09

Control Algorithms

Motion Control: PID + Feedforward

Each joint uses a cascaded control structure: an outer position loop, an intermediate velocity loop, and an inner current (torque) loop. Feedforward terms (velocity, acceleration, gravity compensation) improve tracking accuracy and reduce following error during fast trajectories.

Trajectory Planning

Joint-space trajectories are generated as quintic (5th-order) polynomial splines, ensuring continuous position, velocity, and acceleration profiles. Cartesian linear and circular interpolation is supported for torch-path segments. Velocity blending at waypoints maintains TCP speed continuity using the CNT (continuous-path) blend algorithm.

Adaptive Arc Control

A model-reference adaptive controller (MRAC) maintains arc length in the presence of workpiece surface variation. The reference model is a first-order system with 50 ms time constant. Adaptation gain is scheduled based on wire feed speed to prevent instability at high deposition rates.

Seam Tracking Algorithm

The vision sensor delivers joint lateral offset and height correction at 500 Hz. A Kalman filter fuses sensor data with the nominal programmed path to reject measurement noise and arc-radiation interference. Corrections are applied with a 1-cycle (1 ms) delay via TCP offset commands to the motion controller.

Algorithm	Type	Update Rate	Bandwidth
Position loop	PD + feedforward	1 kHz	100 Hz
Velocity loop	PI	1 kHz	300 Hz
Current loop	PI + decoupling	8 kHz (PWM)	2 kHz
Arc length control	Adaptive PI (MRAC)	10 kHz	20 Hz
Seam tracking (lateral)	Kalman + PID	500 Hz	50 Hz
Seam tracking (height)	Kalman + PID	500 Hz	50 Hz

CHAPTER 10

CAD & Design

3D Model Library

Native CAD files are provided in STEP AP214 and Parasolid (.x_t) formats, compatible with all major CAD platforms (CATIA, NX, SolidWorks, Inventor, Creo). The model library includes the robot arm, controller cabinet, welding torch, wire feeder, cooling unit, and mounting fixtures.

- Robot arm assembly — 1 450 parts, fully constrained
- Motion simulation model — joint limits, collision meshes, TCP frames
- Simplified model for offline programming — 120-face mesh per link
- Mounting template — base plate footprint, anchor bolt pattern

Bill of Materials (BOM)

Item No.	Description	Qty	Part Number	Supplier
1	Servo Motor J1/J2 (1.5 kW)	2	SGM7J-15A	Yaskawa
2	Servo Motor J3 (750 W)	1	SGM7J-08A	Yaskawa
3	Servo Motor J4/J5 (400 W)	2	SGM7J-04A	Yaskawa
4	Servo Motor J6 (200 W)	1	SGM7J-02A	Yaskawa
5	Harmonic Drive J1 (1:80)	1	SHD-32-80	Harmonic Drive AG
6	Laser Vision Sensor	1	Scansonic ALO3	Scansonic
7	EtherCAT Servo Drive	6	SERVOPACK SGD7S	Yaskawa
8	Welding Power Source 500 A	1	TPS 500i	Fronius

CHAPTER 11

Electronics

Main Control PCB

The main control board (MCB-01) hosts the motion controller CPU, EtherCAT ASIC (ET1100), FPGA (Xilinx Artix-7) for I/O processing, and DDR4 RAM. The board is a 6-layer PCB (FR4, 1.6 mm) designed to IPC-A-600 Class 3. Conformal coating (Humiseal 1A33) provides protection against humidity and conductive contamination.

Signal Conditioning

All analog sensor signals (arc voltage, current, F/T sensor) pass through 4th-order anti-aliasing filters ($f_c = 3$ kHz) before 16-bit ADC conversion. Differential input amplifiers (INA826) with ± 2 kV common-mode rejection protect against welding-arc-induced transients. Signal grounds are isolated from power grounds by 1 M Ω / 1 nF networks.

I/O Specifications

I/O Type	Count	Specification
Digital Inputs (24 V)	32	PNP/NPN selectable, 5 ms filter
Digital Outputs (24 V)	32	Source, 500 mA each, short-circuit protected
Analog Inputs	8	± 10 V or 4–20 mA, 16-bit, 10 kHz
Analog Outputs	4	± 10 V, 16-bit, 10 kHz
Encoder Inputs	6	RS422 differential, 4 MHz max
Safety I/O (dual-channel)	8+8	PLe Cat 4, 8 ms scan

CHAPTER 12

Safety & Standards

Applicable Standards

Standard	Title	Compliance Level
ISO 10218-1:2011	Robots and robotic devices — Safety requirements (robot)	Full
ISO 10218-2:2011	Safety requirements (robot system and integration)	Full
ISO/TS 15066:2016	Collaborative operation safety requirements	Applicable clauses
IEC 62061:2021	Safety of machinery — Functional safety (SIL)	SIL 2 (control system)
ISO 13849-1:2015	Safety of machinery — Safety-related parts (PL)	PLd (safety functions)
IEC 60974-1:2017	Arc welding equipment — Welding power sources	Full
EN ISO 11553-1	Safety of laser processing machines	Class 2 laser sensor
2006/42/EC	EU Machinery Directive	CE marked

Safety Functions

- STO — Safe Torque Off: removes drive enable; PLe/SIL3 via dual channel
- SLS — Safe Limited Speed: max 250 mm/s in reduced speed mode
- SLP — Safe Limited Position: joint range monitoring in safety controller
- SBC — Safe Brake Control: holds brakes on power loss
- Collision detection: current-based < 5 ms, triggers protective stop
- Workspace monitoring: area scanner enforces safety zones

Risk Assessment Summary

A risk assessment per ISO 12100 identifies 14 significant hazards. Residual risk after safety measures is "Low" for all hazards. Key measures include physical guarding (fencing + light curtains), safe-rated motion monitoring, operator training, and annual safety function verification (proof test interval: 1 year).

CHAPTER 13

Manufacturing

Recommended Welding Processes

Process	Base Material	Wire/Filler	Shielding Gas	Current (A)
MIG/MAG (GMAW)	Mild steel	ER70S-6 Ø1.2 mm	Ar+18%CO2	150–350
MIG/MAG (GMAW)	Stainless 316L	ER316L Ø1.0 mm	Ar+2%CO2	100–250
MIG/MAG (GMAW)	Aluminium 5083	ER5356 Ø1.2 mm	100% Ar	120–280
TIG (GTAW)	Stainless / Ti	ER308L Ø2.4 mm	100% Ar	60–180
Flux-Core (FCAW)	Mild/HSLA steel	E71T-1C Ø1.2 mm	CO2 100%	200–450
Plasma Arc	Thin stainless	None (autogenous)	Ar + H2 5%	30–120

Weld Quality & Inspection

- Visual inspection per ISO 5817 Level B after each production run
- Automated bead geometry monitoring via laser profilometer (post-weld)
- Magnetic particle testing (MT) for ferromagnetic assemblies — monthly audit
- Radiographic or ultrasonic testing for pressure-vessel applications
- Statistical process control (SPC) charts for voltage, current, speed
- Weld log stored per job: date, operator, parameters, sensor data

Maintenance Schedule

Interval	Task	Responsible
Daily	Clean nozzle, check wire feed, inspect torch	Operator
Weekly	Lubricate liner, check water level, inspect cables	Maintenance
Monthly	Check TCP accuracy, clean optics, log brake wear	Maintenance
Quarterly	Drive firmware backup, safety function test	Specialist
Annually	Full safety system proof test, gearbox oil (if applicable)	OEM Service
20 000 h	Bearing replacement, harmonic drive inspection	OEM Service

CHAPTER 14

Research Papers & References

Key Academic References

Author(s)	Title	Publisher/Journal	Year
Craig, J.J.	Introduction to Robotics: Mechanics and Control, 3rd ed.	Pearson	2005
Spong, M.W., Hutchinson, S., Vidyasagar, M.	Robot Modeling and Control, 2nd ed.	Wiley	2020
Niku, S.B.	Introduction to Robotics: Analysis, Control, Applications	Wiley-IEEE	2020
Siciliano, B. et al.	Robotics: Modelling, Planning and Control	Springer	2009
Lincoln, D. et al.	Adaptive Seam Tracking for Robotic GMAW	Welding Journal	2021
Zhang, Y. et al.	Real-time Arc Sensing for GTAW Process Control	Journal of Manufacturing Science	2019
Lee, S. et al.	Machine Learning for Weld Defect Detection	NDT & E International	2022
Kim, I.S. et al.	A Study on Application of Adaptive Control to Arc Welding	Int. J. Adv. Manufacturing Tech.	2018

Patent Citations

- US10456956B2 — Laser vision seam tracking system with adaptive gain
- EP3412397A1 — Collaborative robot welding cell with force-limited motion
- JP6457469B2 — Arc-length adaptive control using MRAC for GMAW
- CN110587600A — Multi-layer weld path planning for thick-section joints

CHAPTER 15

Datasheets & Component Specifications

Key Component Summary

Component	Manufacturer	Part Number	Datasheet Rev.	Key Spec
Servo Drive	Yaskawa	SGD7S-2R8A	Rev. 5	2.8 A, 400 V, EtherCAT
Servo Motor	Yaskawa	SGM7J-15A	Rev. 3	1.5 kW, 3000 rpm, 23-bit enc
Welding Power Source	Fronius	TPS 500i	Rev. C	500 A, 100 % DC, CMT capable
Laser Sensor	Scansonic	ALO3	Rev. B	±0.05 mm Z, 500 Hz
Safety PLC	Pilz	PSS 4000-S	Rev. 6	SIL 3 / PLe, EtherCAT FSoE
Area Scanner	SICK	S300 Expert	Rev. 4	3 m range, PLd Cat 3
F/T Sensor	ATI	Omega-160-IP65	Rev. A	±1250 N, ±50 N·m, EtherCAT
EtherCAT I/O	Beckhoff	EK1100 / EL2008	Rev. 7	8 DO, 24V, 0.5A each

Full component datasheets are stored in the "15_Datasheets" folder of the project repository. Revision control is managed in the company PLM system (Teamcenter). Always verify the latest revision before ordering replacement parts.

CHAPTER 16

Metadata & Document Control

Document History

Rev.	Date	Author	Approved By	Changes
A	2024-03-15	R. Sharma	D. Patel	Initial release
B	2024-07-22	R. Sharma	D. Patel	Added seam tracking chapter
C	2024-11-08	A. Kumar	D. Patel	Updated safety standards (ISO 10218:2024 draft)
2.0	2025-06-01	A. Kumar / R. Sharma	D. Patel	Full restructure; added kinematics, algorithms

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